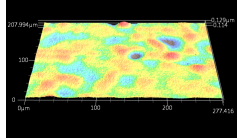
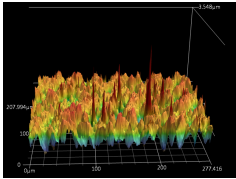


GHZ HIS Carrier Copper Foil Comparison Datasheet			
LongYoung Electronics (GHZ) Nanometer-Scale Surface Control & 3D Micro-Topography Measurement Data			
I. Production Process			
Comparison Dimension	Product Description		Production Site
HIS HIS1: Rz 0.8um, L/S 10/10um HIS2: Rz 0.5um, L/S 10/10um or less	GHZ's pioneering vacuum sputtering and precision plating technology overcomes the limits of traditional process, dramatically lowering surface roughness for high-frequency applications while securing optimal peel strength. Designed for multi-cycle thermal lamination, the advanced release layer guarantees a highly stable release force, ensuring easy and clean carrier peeling under harsh high-temperature laminating conditions.		 
II. Core Product Technical Metrics Comparison			
Comparison Dimension	GHZ	Mitsui	Materials Analysis & Superiority Comparison
Product Copper Thickness	1.5 / 2 / 3 μm	1.5 / 2 / 3 μm	Traditional copper foils are prone to pinhole defects at ultra-thin levels due to high roughness; GHZ stably delivers a 1.5μm ultra-thin defect-free continuous copper layer.
Target L/S by MSAP	HIS1: Rz 0.8um, L/S 10/10um HIS2: Rz 0.5um, L/S 10/10um or less	MT18FL: Rz 1.3um, L/S 15/15um MT18GN: Rz 0.9um, L/S 10/10um	Mitsui (MT18GN) Requires reduction of roughness to Rz 0.9 μm to prevent CAF or residual copper. GHZ Advantage (HIS2): Pushes the roughness down to Rz 0.5 μm. This ultra-low profile dramatically widens the flash etching process window and eliminates the risk of undercut or necking, serving as the gold standard for sub-10 μm circuitry.
Surface Roughness (Ra)	< 0.02 μm (Typical: 0.006 μm)	> 0.1 μm (Typical: 0.187 μm)	Skin Effect: Ra differs by nearly 30 times. Under high-frequency transmission, current flows only on the surface. GHZ reduces insertion loss by 15%+.
Ten-Point Height (Rz)	< 0.5 μm (Typical: 0.039 μm)	0.9 ~ 2.0 μm (Typical: 1.115 μm)	Insulation Performance: Deep roots of traditional foils pierce resin, causing CAF micro-shortages. GHZ has no roots, securing insulation for 10μm fine lines.
Effective Copper Thickness	No anchor roots, highly uniform thickness	Highly fluctuating due to varying root depth	Traditional thickness variation hurts impedance control. GHZ's extremely uniform thickness ensures precise characteristic impedance control.
Crystal Structure	Nanometer-scale grains, fine and uniform	Coarse columnar grains	Fine equiaxed grains ensure identical etching rates in all directions, delivering high etch factor and vertical sidewalls without undercut (no side-etching).
Flash Etching Controllability	Excellent (Wide process window)	Unstable (Highly prone to over-etching)	During MSAP flash etching (carrier copper removal), GHZ's uniform 1.5μm copper layer strips rapidly and uniformly, preventing fine-line necking or breakage.
Thickness Uniformity	Extremely flat (Mirror-like finish)	Unstable	Mirror-like bonding interface effectively prevents mechanical thermal stress mismatch during lamination, eliminating substrate warpage risks.
III. 3D Micro-Topography Measurement Statistics (White Light Interferometry)			
3D Measurement	GHZ (HIS2)	Mitsui (MT18FL)	Parameter Analysis & Superiority Ratio
			GHZ HIS Series Carrier Copper Foils - Characterized by high thickness uniformity, molecular-level flatness, and a completely anchor-tooth-free bonding interface. - Capable of delivering stable, defect-free continuous copper layers down to 1.5 μm. - Precisely engineered for sub-10 μm ultra-fine line circuitry (mSAP) and high-density semiconductor packaging substrates.
Ra	0.006 μm (6 nm)	0.187 μm (187 nm)	Competitor roughness is 31 times that of GHZ. GHZ achieves nanometer mirror finish, minimizing skin-effect insertion loss.
Rz	0.039 μm (39 nm)	1.115 μm (1115 nm)	Competitor root fluctuations are 28.6 times that of GHZ. GHZ eliminates deep spikes piercing the resin, ensuring high insulation resistance.
Sa	0.013 μm	0.249 μm	Evaluates overall 3D area fluctuations. Competitor regional roughness is 19.1 times that of GHZ.
Sz	0.217 μm	4.757 μm	Represents the deepest anchor tooth height. Competitor maximum peak is 22 times that of GHZ, posing a severe risk of residual copper.
Sdr	0.0001044	0.0478500	Represents additional surface area ratio from micro-roughness. Competitor area increase ratio is 458 times that of GHZ due to dense roots.
Spc	61.415 1/mm	2074.628 1/mm	Represents micro-spike sharpness. Competitor peak curvature is 33 times that of GHZ, where electric fields easily concentrate and cause dielectric breakdown.

Should you have any questions or require further details, we would be pleased to arrange a technical seminar at your convenience.